

together with Alexey Chervonenkis, he proposed the Vapnik–Chervonenkis (VC) dimension, a theoretical framework to measure the capacity of a classification model.

Now, consider the program shown in Figure 3. First, let us try to understand the code. The program uses SVM to classify data. Line 1 imports the `svm` module from scikit-learn. Notice that scikit-learn can be very easily installed by using Anaconda Navigator, as shown in the previous articles in this series. Line 2 defines a list called `X` which contains the training data. All the elements in `X` are lists of size 3. Line 3 defines a list `y`, which contains the class labels associated with the data in the list `X`. In this example, the data is classified into just two classes labelled 0 and 1. However, it is possible to classify data into multiple classes using the same technique. Line 4 uses the `SVC()` method provided by the module `svm`. `SVC()` method generates a support vector classifier. Line 5 uses the `fit()` method provided by the module `svm` to fit the SVM classifier model according to the given training data, in this case the arrays `X` and `y`. Finally,

```
1 from sklearn import svm
2 X = [[1,1,1], [2,2,2], [3,33,333], [4,44,444]]
3 y = [0,0,1,1]
4 clf = svm.SVC()
5 clf.fit(X, y)
6 print(clf.predict([[5., 5., 5.]])
7 print(clf.predict([[5., 55., 555.]])
```

[0]
[1]

Figure 3: Classification with SVM

```
1 from sklearn import svm
2 X = [[1,1], [2,2], [3,3], [4,4], [5,5]]
3 y = [100.0, 200.0, 300.0, 400.0, 500.0]
4 regr = svm.SVR()
5 regr.fit(X, y)
6 print(regr.predict([[1, 4]]))
```

[299.72775944]

Figure 4: Regression with SVM

lines 6 and 7 make predictions based on this classifier. The results of the predictions are also shown in Figure 3. Notice that the classifier is able to distinguish between the two sets of data we have provided for testing.

Now, let us see an example where SVM is used for regression. Consider the code shown in Figure 4. First, let us try to understand the code. Here again, Line 1 imports the `svm` module from scikit-learn. Line 2 defines a list called

`X`, which contains the training data. Notice that all the elements in `X` are lists of size 2. Line 3 defines a list `y` which contains the values associated with the data in the list `X`. Line 4 uses the `SVR()` method provided by the module `svm`. This method generates a support vector regression model. Line 5 uses the `fit()` method provided by the module `svm` to fit the SVM regression model according to

the given training data, in this case the arrays `X` and `y`. Finally, Line 6 makes a prediction based on this SVM regression model. The result of this prediction is also shown in Figure 4. There are two other methods in addition to `SVR()`, `LinearSVR()` and `NuSVR()`, to implement a support vector regression model. Replace Line 4 with the lines of code '`regr = svm.LinearSVR()`' and '`regr = svm.NuSVR()`', and execute the code to

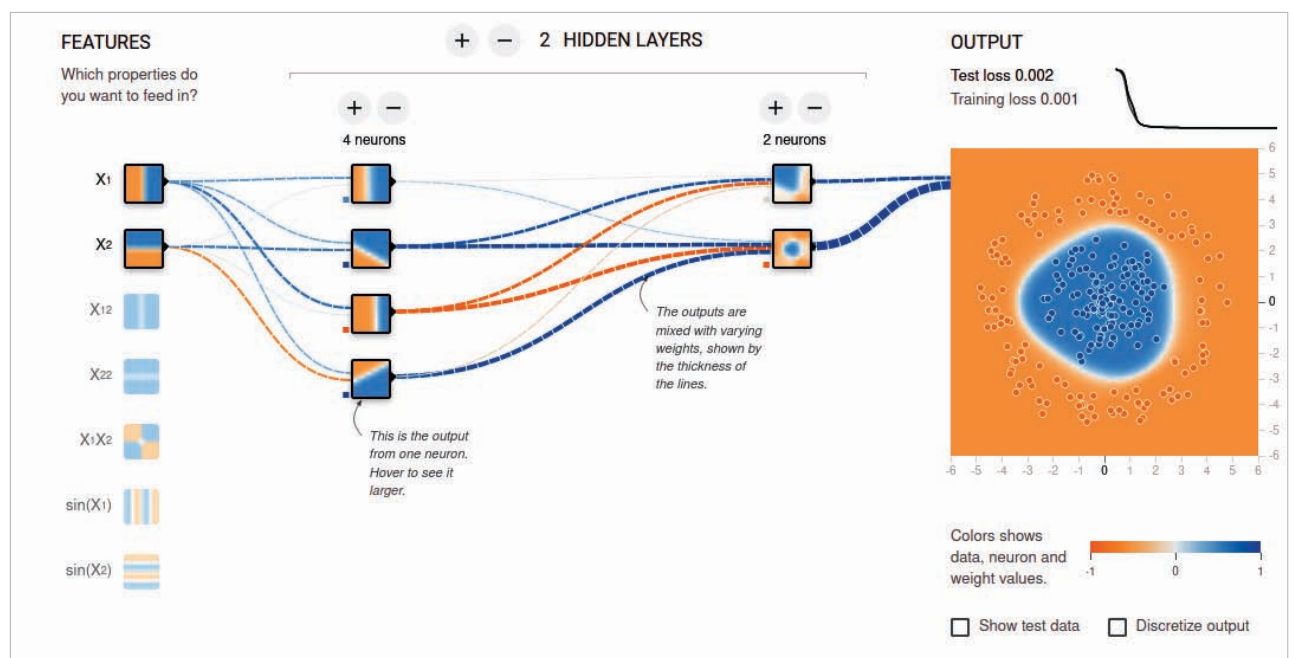


Figure 5: The neural network playground in action